

Abdullah Mahmoud Abdullah

Data Engineer

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PROFESSIONAL SUMMARY

Junior Data Engineer focusing on scalable ETL pipelines, cloud integration, and AI-driven data systems. Strong technical skills in managing large datasets, Collaborative communicator adept at leading technical syncs and aligning engineering efforts with core business objectives.

SKILLS

- Programming & Scripting:** Python (Pandas), SQL, Bash/Shell Scripting
- Cloud Platforms:** AWS Solutions Architecture (S3, Lambda, Glue, Athena, EC2, EventBridge, SNS), Snowflake
- Data Engineering & Big Data:** ETL/ELT Pipelines, Apache Spark, Hadoop, Apache Airflow, Apache Kafka
- Databases:** OLTP (PostgreSQL, RDBMS), OLAP & Data Warehousing, NoSQL
- Business Intelligence & Visualization:** Looker Studio, Streamlit, IBM Cognos Analytics
- Machine Learning & AI:** Generative AI (Google Gemini, Groq API), Predictive Modelling, NLP (Whisper V3)
- Soft Skills:** Structured Problem Solving

EXPERIENCE

Roto House | AI Engineer | Remote

Oct 2025 – Present

- Developed a factory management web app with an AI-driven backend (Node.js, PostgreSQL for OLTP) to automate reporting and digitize industrial workflows, streamlining operations for 70+ active technicians.
- Processed 20+ daily audio using FFmpeg and Whisper V3 (via Groq API) for accurate speech-to-text transcription.
- Reduced manual data entry time by 60% by leveraging Google Gemini to analyze visual workflows and extract key operational data, successfully auto-populating 12+ target forms daily.

PROJECTS

End-to-End Data Engineering Platform (IBM Capstone)

- Architected OLTP (relational) and NoSQL databases to handle diverse data types securely and efficiently.
- Built and orchestrated scalable ETL pipelines utilizing Python, Shell Scripting, and Apache Airflow.
- Leveraged Apache Spark to process datasets, integrating the data warehouse with Looker Studio for interactive dashboards.

Serverless Real-Time Crypto Data Pipeline

- Engineered an event-driven AWS pipeline (SQS, Lambda, DynamoDB) to process live Binance WebSocket data.
- Provisioned infrastructure and implemented real-time automated SNS alerts for micro-batch market volatility.
- Built a Streamlit dashboard integrating Gemini 3-flash-preview for live quantitative market analysis.

Orchestrating an IMDb ELT Pipeline: AWS to Snowflake Integration

- Built an automated AWS ELT pipeline using EventBridge, Lambda, and SNS to extract over 670 MB IMDb datasets.
- Optimized Snowflake OLAP ingestion using CDC and SQL MERGE, eliminating full-loads to achieve high data availability.
- Developed a Streamlit dashboard powered by Snowflake, enabling live data visualization via SQL pushdown.

YouTube AWS Serverless Data Pipeline

- Engineered a data pipeline, utilizing AWS Lambda for the extraction of raw JSON and CSV files from Amazon S3.
- Processed and optimized datasets using AWS Glue by partitioning data and converting files to Parquet format.
- Centralized the transformed data within Snowflake and developed dashboards using Looker Studio and Streamlit.

Global Weather Analytics Pipeline

- Engineered an automated Python and Airflow ETL pipeline to extract daily weather data from the Open-Meteo API.
- Transformed complex, nested JSON payloads into structured, analysis-ready datasets utilizing Pandas.
- Designed an interactive Streamlit dashboard for dynamic visualization and analysis of global weather trends.

CERTIFICATIONS

- IBM Data Engineering Professional Certificate** | Coursera | Mar 2026
- Snowflake Data Engineering Professional Certificate** | Coursera | Apr 2026
- AWS Cloud Solutions Architect** | Coursera | May 2026

EDUCATION

Bachelor of Computer Science | Faculty of Computers and Artificial Intelligence | Sep 2022 - Jul 2026

- Relevant Coursework:** Data Modeling, Data Architecture, Database Management, Artificial Intelligence.